raffle.fun

TECHNICAL REFERENCE

raffle.fun: A Technical Whitepaper

State machine, economics, invariants, threat model and integration surface for an immutable, administrator-free NFT raffle protocol.

READER

Auditors, protocol engineers, integrators

PROTOCOL COMMIT

e65e1e5f89a5

NOT DEPLOYED

NOT INDEPENDENTLY AUDITED

Pre-release

An immutable, administrator-free NFT raffle protocol with constant-time range tickets, bearer settlement, and hard-bounded oracle liveness.

FIELD	VALUE
Version	Contracts `1.0.0`
Protocol commit	`e65e1e5f89a548ed03a0ebe0a0b722d609244d18` (production Solidity)
Document date	2026-08-20
Audience	Security auditors, protocol engineers, researchers, integrators, sophisticated participants
Target chain	Ethereum mainnet (1); Sepolia (11155111) required first
Compiler	Solidity `0.8.36`, exact pragma, EVM target `cancun`, optimizer enabled at 200 runs, via-IR disabled
Dependencies	`@openzeppelin/contracts@5.6.1`, `@chainlink/contracts@1.5.0`
Deployment status	**Not deployed.** No deployment record exists for any network.
Audit status	**Not independently audited.**

0. Scope, precedence, and non-claims

0.1 Precedence

This document describes the production Solidity in `packages/contracts/src/` at the commit above. Where any other artifact disagrees — including this document — **the deployed bytecode is authoritative for onchain behavior**.

Every substantive claim below is backed by a Fact ID in `docs/facts/raffle-fun-facts.md`, which cites the contract, line, function, and test for each. Fact IDs appear inline as `[RF-nnn]`.

0.2 Evidence of record

`packages/contracts/audit/CURRENT-*.md` is the internal evidence set for this candidate. `V12-REVIEW-2026-08-20.md` is an independent review of a supplied third-party finding export at commit `3da958f`; it promoted none of its fifteen entries to a confirmed in-scope exploit and states explicitly that it "is not a mainnet go-ahead" `[RF-066]`.

A freshness caveat applies throughout §13: the recorded campaign totals were captured at implementation SHA `92eccb4`, before the hard request/callback-boundary remediation, the official Chainlink consumer-base migration, the bearer-redemption redesign, and the ownerless-factory change. Behavioral statements in this document are read from current source; the **numeric** totals are preserved evidence for earlier SHAs and must be reproduced from a clean checkout of the release SHA.

0.3 Superseded artifacts

Several documents in this repository describe an **earlier** architecture — a Base deployment, Pyth Entropy randomness, a read-aggregator Lens, an `Ownable2Step` factory, per-ticket minting, and a variable ticket price — and must not be relied on:

- `docs/whitepaper/**` (the retired generation's source, build pipeline, fact-check register, and pre-generated SVG figures). Its fact generator still requires symbols that no longer exist in production Solidity, so `pnpm docs:whitepaper` cannot run.
- `packages/contracts/audit/` reports dated 2026-08-13 or earlier, plus `INTERNAL-AUDIT.md`, `INDEPENDENT-SPECIFICATION.md`, `FINDINGS.md`, `MUTATION-TESTING.md`, `TEST-CAMPAIGN.md`, `RESIDUAL-RISKS.md`, and `RELEASE-READINESS-2026-08-17.md`.

§16.2 tabulates precisely what was removed, so a future writer cannot reintroduce retired behavior from a historical report.

0.4 Non-claims

This document does **not** assert that raffle.fun is audited, formally verified, trustless, provably fair, risk-free, guaranteed, live, or production-ready. The project's own campaign record states the disposition verbatim: internally audit-ready for independent review; **not mainnet-ready** `[RF-064]`. Nine release-verification gaps (`V1-REL-01` … `V1-REL-09`) remain open.

0.5 Security objective (as stated by the project)

> For an honest standards-compliant ERC-721 prize and available exact-transfer official USDC, no unauthorized party can select the winner, seize or duplicate a supported asset, create an unfunded liability, redirect a fixed recipient's balance, or make work scale with total entries. A sold raffle has a bounded two-day request window followed, if a request succeeds, by a bounded two-day callback window, each with an exact full-refund recovery boundary — subject to external asset and chain availability.

The objective explicitly excludes lost keys, contracts incapable of acting, malicious or upgraded token code, issuer freezes and blocklists, burned escrowed NFTs, dishonest token reads, chain halt, universal censorship, and unrelated assets forced into the contract. See §12 and §13.

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1. Design goals and rationale

raffle.fun targets a narrow problem: run a single-prize NFT raffle onchain such that no party — including the protocol operator — can alter the outcome, withhold settlement, or strand assets, while keeping every execution path bounded in gas **independently of how much was sold**.

Six design decisions follow from that and shape everything else.

G1 — No administrator anywhere in the system. Neither the factory nor any raffle has an owner, role, pause, upgrade, rescue, or mutable configuration `[RF-001]`, `[RF-008]`. Every transition is gated only by lifecycle status, `block.timestamp`, ERC-721 ownership, and wrapper authentication. This eliminates governance and key-compromise risk entirely, at the cost of eliminating all remediation `[RF-009]`.

G2 — One implementation, many clones. Each factory deploys one locked `Raffle` implementation and creates fixed-target ERC-1167 minimal proxies `[RF-006]`. Every canonical raffle provably runs identical code; there is no proxy admin, beacon, CREATE2 salt, or upgrade path. The clone's 45-byte runtime hard-codes its implementation, so "immutable" needs no argument beyond reading the factory constructor.

G3 — Range tickets, not per-entry tokens. One purchase mints one ERC-721 holding an inclusive `[firstEntry,lastEntry]` `uint128` range `[RF-017]`. Buying one entry and buying a million entries cost the same in mints and storage writes. This is what makes purchase, resolution, and winner proof O(1) in entry count `[RF-063]`.

G4 — Bearer credentials, and settlement that never reads ownership. `settleWinningTicket` proves which ticket contains the winning entry and allocates every liability without calling `ownerOf`, burning, or transferring `[RF-041]`. The ticket stays an ordinary transferable ERC-721 — in *every* status, including after the winner is known — until its current owner burns it to redeem `[RF-021]`. Secondary markets work natively; destination risk moves entirely onto the user.

G5 — Hard-bounded liveness with terminal fallback. Both ways the protocol can stall have an exact deadline, after which a **permissionless** function converts the raffle to full, fee-free refunds `[RF-050]`. Requests and

callbacks are excluded *at* their deadlines and refunds open *at* them, so the two are mutually exclusive by construction rather than by race `[RF-028]`, `[RF-029]`.

G6 — Fixed recipients and exact-delta verification. No caller can name a payout destination: value goes to the current ticket owner, the immutable `sponsorRecipient`, or the immutable `protocolTreasury` `[RF-044]`. Both incoming and outgoing ERC-20 movements are verified by two-sided balance delta `[RF-016]`, `[RF-055]`, and NFT delivery is verified by `ownerOf` post-condition `[RF-043]`.

1.1 Explicit non-goals

- Multi-prize, multi-winner, or tiered raffles.
- Multiple quote tokens, or a variable entry price, per factory.
- Any form of privacy `[RF-077]`.
- Reroll, second oracle, or randomness fallback `[RF-059]`.
- Any onchain economic value ceiling `[RF-073]`.
- Cross-raffle asset recovery. This was implemented once, found exploitable, and deliberately removed (§16.2).

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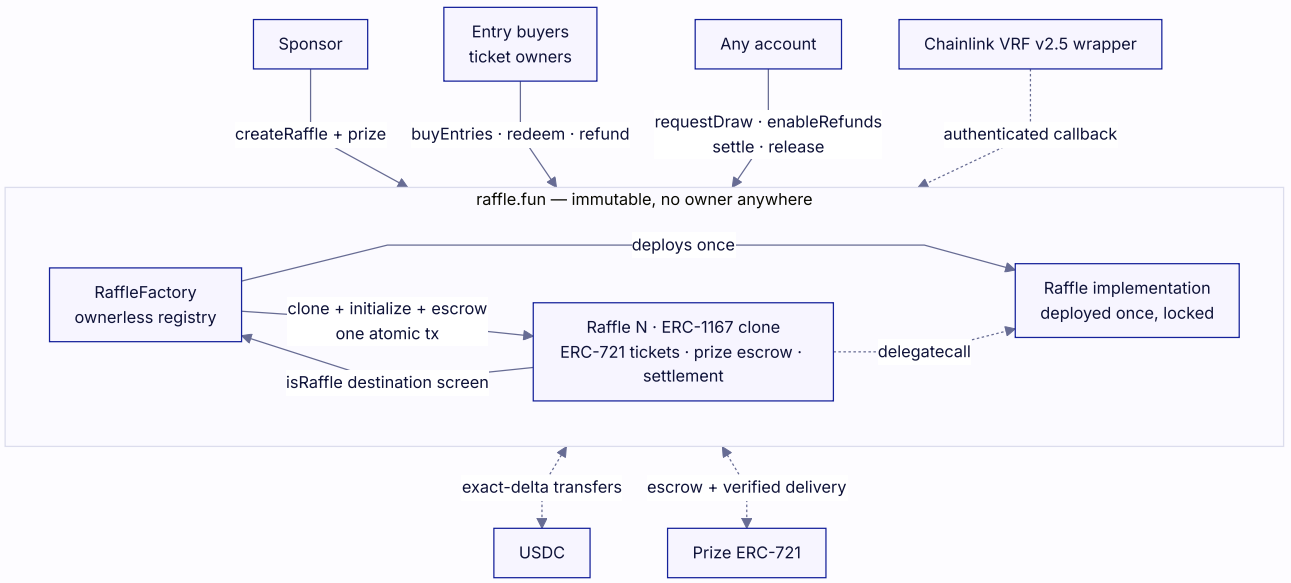
2. System architecture



Two production contracts, no privileged offchain component, and every arrow crossing the dashed boundary is an external dependency the protocol trusts but cannot control.

There are exactly **two** production contracts and no offchain privileged component. There is no read-aggregator Lens: integrators read raffles directly and authenticate them through the factory registry (§14.1).

2.1 Contract responsibilities

CONTRACT	HOLDS ASSETS	MUTABLE STATE	AUTHORITY
`RaffleFactory`	No	Registry mappings and `raffleCount` only	**None** `[RF-001]`
`Raffle` (clone)	Yes — one ERC-721 prize + quote-token liabilities	Lifecycle, liabilities, ERC-721 ledger	**None** `[RF-008]`

Both rows read "None". That is the whole point of the design, and it is checked as an ABI-surface property, not merely asserted.

2.2 Inheritance

```
Raffle        is IRaffle, ERC721, ReentrancyGuard, IERC721Receiver, VRFV2PlusWrapperConsumerBase
RaffleFactory is IRaffleFactory, ReentrancyGuard
```

`VRFV2PlusWrapperConsumerBase`, `IVRFV2PlusWrapper`, and `VRFV2PlusClient` come from the exact-pinned official `@chainlink/contracts@1.5.0` `[RF-034]`. The protocol does not reimplement a coordinator or wrapper, and native direct funding means no raffle creates or manages a VRF subscription.

2.3 Bytecode footprint

Measured with `forge build --sizes` at this document's commit `[RF-062]`:

CONTRACT	RUNTIME (B)	INITCODE (B)	RUNTIME MARGIN (B)	INITCODE MARGIN (B)
`Raffle`	17,459	18,569	7,117	30,583
`RaffleFactory`	3,973	23,476	20,603	25,676

Runtime margin is against EIP-170 (24,576 B); initcode margin against EIP-3860 (49,152 B). The factory's runtime is small because the protocol logic lives in the cloned implementation; its **initcode** is large because its constructor deploys that implementation.

> **Auditor note.** The retired generation carried a 309-byte EIP-170 margin on the factory and treated the size gate as release-sensitive. The clone architecture removed that constraint. Re-measure on the frozen release SHA anyway.

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3. Formal lifecycle

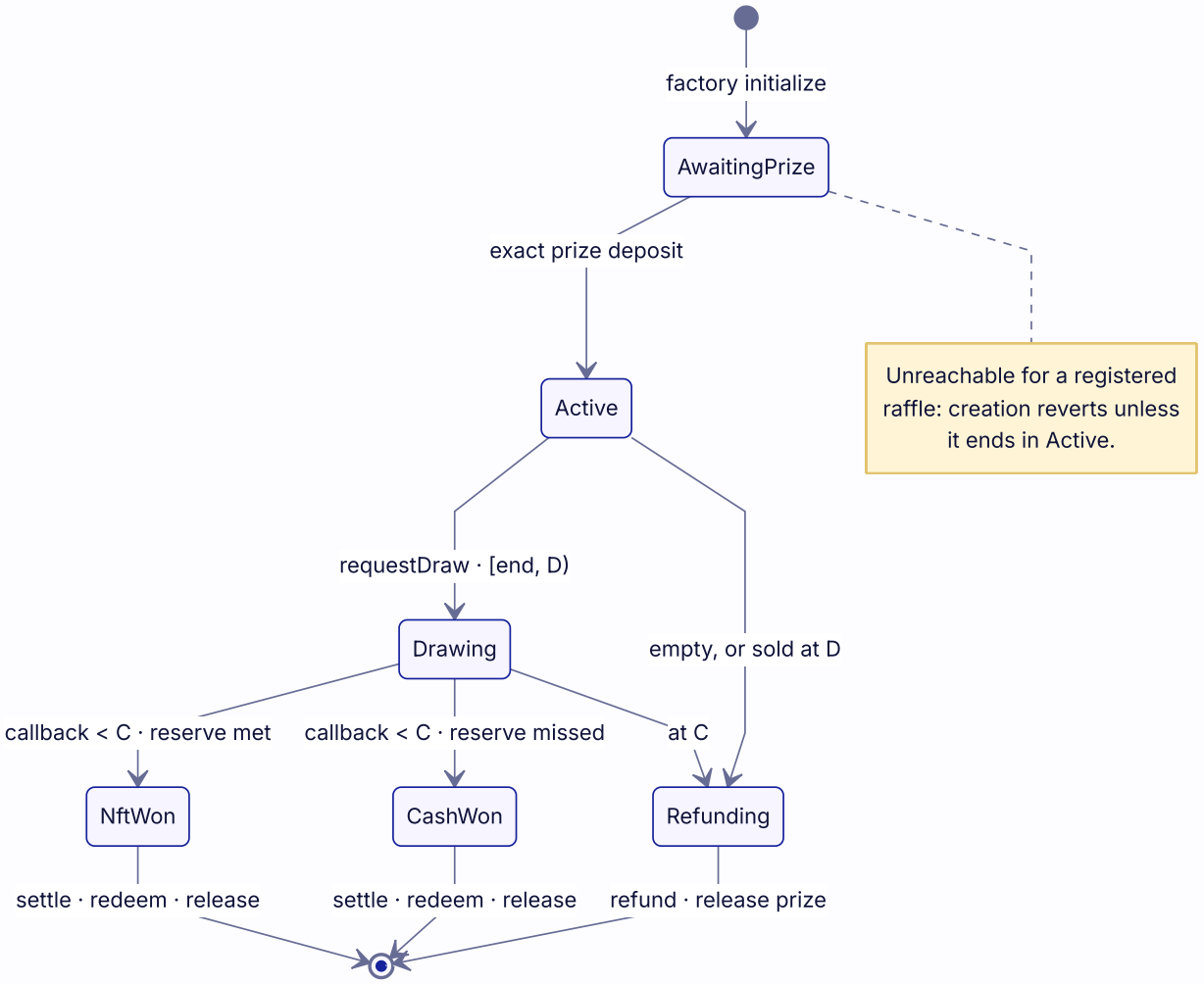
3.1 The `Status` enum

`IRaffle.Status` is the sole lifecycle **and** economic-outcome representation. There is no second outcome enum and no `Closed` state.

ORDINAL	NAME	MEANING
0	`AwaitingPrize`	Exists only inside the creation transaction `[RF-011]`
1	`Active`	Prize escrowed; sale open while `now < endTime`
2	`Drawing`	Exactly one VRF request pending
3	`NftWon`	Reserve met; prize reserved for the winning ticket
4	`CashWon`	Reserve missed; prize returns to the sponsor side
5	`Refunding`	A liveness deadline expired, or the raffle sold nothing

> **Auditor note.** Ordinal `0` is unreachable for any *registered* raffle: creation's post-condition requires `Active`, so a raffle that fails to activate is never registered `[RF-011]`. It is, however, the live status of the locked implementation, which the constructor parks in `Refunding` with zero liability `[RF-007]`.

3.2 Transition graph



The complete lifecycle. All three terminal states are absorbing — unlike the retired design, NftWon has no edge back to Refunding, so a resolved result is final.

`NftWon`, `CashWon`, and `Refunding` are **absorbing**. Unlike the retired design, `NftWon` has no outbound edge to `Refunding`: a resolved result is final `[RF-040]`. §9.5 states the consequence without softening.

3.3 Transition guards

Let `D = drawRequestDeadline() = endTime + 2 days` and `C = callbackDeadline() = drawRequestedAt + 2 days`.

FROM → TO	TRIGGER	GUARDS
`—` → `AwaitingPrize`	`initialize`	`!initialized` ∧ `msg.sender == factory` ∧ no zero party ∧ no party is a known protocol destination `[RF-007]`
`AwaitingPrize` → `Active`	`onERC721Received`	`msg.sender == prizeToken` ∧ `tokenId == prizeTokenId` ∧ `from == sponsor` ∧ `operator == factory` `[RF-012]`
`Active` → `Drawing`	`requestDraw`	`now >= endTime` ∧ `totalEntries != 0` ∧ `now < D` ∧ `msg.value >= quotedFee` `[RF-028]`, `[RF-030]`
`Active` → `Refunding`	`enableRefunds`	`totalEntries == 0` ∧ (`msg.sender == sponsor` ∨ `now >= endTime`) — **or** — `totalEntries != 0` ∧ `now >= D`
`Drawing` → `NftWon`	`fulfillRandomWords`	wrapper-authenticated ∧ `!_requestInFlight` ∧ `requestId == vrfRequestId` ∧ one word ∧ `now < C` ∧ `totalEntries >= reserveEntries`

FROM → TO	TRIGGER	GUARDS
`Drawing` → `CashWon`	`fulfillRandomWords`	same, with `totalEntries < reserveEntries`
`Drawing` → `Refunding`	`enableRefunds`	`now >= C`

Any other `enableRefunds` status reverts `InvalidStatus`; before the applicable deadline it reverts `RefundsNotAvailable(deadline, now)`.

3.4 Timing envelope

All constants from `RaffleConstants.sol` `[RF-027]`, `[RF-028]`, `[RF-029]`:

CONSTANT	VALUE
`MAX_SALE_DURATION`	30 days
`DRAW_REQUEST_TIMEOUT`	2 days
`DRAW_CALLBACK_TIMEOUT`	2 days
`MAX_REFUND_TICKET_BATCH_SIZE`	100

There is no start delay and no NFT-redemption timeout. The sale begins the moment the prize deposit activates the clone, inside the creation transaction `[RF-015]`.

Worst-case custody bound. From creation, the longest interval before *some* terminal asset path is unconditionally available is:

```
30d (sale) + 2d (request window) + 2d (callback window) = 34 days
```

The almost-four-days subtlety. The two windows are two days each, but they are sequential rather than overlapping, and the second one starts when the request lands. A request included at `D − 1` receives a *full fresh* two-day callback window, so the last nominal boundary after sale end is

```
(2 days − 1 second) + 2 days  ≈  3 days 23:59:59
```

— just under four days, not two. Integrators and monitoring must key on `callbackDeadline()`, never on `endTime + 4 days` as a constant.

The two windows are sequential, not overlapping. A request landing one second before D still earns a full fresh callback window, which is where the almost-four-day tail comes from.

Sale boundaries are **exclusive end**, and the request window opens inclusively at the same instant, so the sale and draw windows abut with no dead interval `[RF-026]`.

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4. Creation and prize escrow

4.1 Atomic construction

`RaffleFactory.createRaffle` performs, in a single transaction `[RF-010]`:

1. `_validateCreateParams` (§4.2).
2. `endTime > block.timestamp` and `endTime - now <= MAX_SALE_DURATION`.
3. `raffleId = ++raffleCount`.
4. `raffle = Clones.clone(raffleImplementation)` — a standard EIP-1167 minimal proxy.
5. `IRaffle(raffle).initialize(...)` with `sponsor = msg.sender`, the factory's immutable `protocolTreasury`, and the caller's `sponsorRecipient`, `prizeToken`, `prizeTokenId`, `reserveEntries`, `endTime`. Status becomes `AwaitingPrize`.
6. Registry writes: `raffleById`, `idByRaffle`, `isRaffle` `[RF-005]`.
7. `prizeToken.safeTransferFrom(msg.sender, raffle, prizeTokenId)` → triggers the clone's receiver hook → status `Active`.
8. **Post-conditions:** `ownerOf(prizeTokenId) == raffle` **and** `IRaffle(raffle).status() == Active`, else `PrizeEscrowVerificationFailed`.
9. `emit RaffleCreated(...)` (ten fields, indexer-complete).

Any failure reverts everything, including the clone deployment and every registry write. **No registered raffle can remain in `AwaitingPrize` `[RF-011]`.**

Auditor note. The double post-condition is deliberate. `ownerOf` alone would not catch a token that invokes the receiver hook without transferring; the status check alone would not catch a token that transfers without invoking the hook. Both together still route through the prize contract, so a fully dishonest ERC-721 defeats both `[RF-056]`. `createRaffle` is `nonReentrant`, and a prize attempting to reenter during escrow is rejected atomically.

Auditor note — initialization ordering. `initialize` writes `initialized = true` *before* validating its parameters. That is safe precisely because creation is atomic: a rejected parameter reverts the whole transaction, so a clone can never be left half-configured but flagged initialized. Slither's fixed-clone initialization heuristic is suppressed at exactly this call with its rationale in source; no detector is disabled globally `[RF-067]`.

4.2 Creation parameter validation

CHECK	FAILURE
`prizeToken.code.length != 0`	`NotContract`
`sponsorRecipient != address(0)`	`ZeroAddress`
`prizeToken` ∉ {factory, quoteToken, vrfWrapper, implementation, any registered raffle}	`UnsafeProtocolDestination`
`sponsorRecipient` ∉ same set ∪ { `prizeToken` }	`UnsafeProtocolDestination`
`supportsInterface(type(IERC721).interfaceId)` returns true; a throwing call is caught	`UnsupportedPrizeToken`
`reserveEntries != 0`	`ZeroReserveEntries`
`endTime > block.timestamp`	`InvalidEndTime`
`endTime − block.timestamp <= 30 days`	`SaleDurationTooLong`

`reserveEntries` is unbounded above. A sponsor may set an unreachable threshold to force the cash branch deterministically — permitted, and frontends should surface it `[RF-038]`.

The clone's own `initialize` re-screens `sponsor`, `sponsorRecipient`, and `protocolTreasury` against `_isInitializationProtocolDestination`, which additionally covers the raffle itself and the implementation `[RF-025]`.

4.3 Receiver authentication

`onERC721Received` reverts with `UnexpectedPrize(token, tokenId, from, operator)` unless all five conditions hold: correct status, correct token contract, correct token ID, sent by the sponsor, operated by the factory `[RF-012]`. Because `AwaitingPrize` is unreachable after activation, a duplicate deposit of the same token is rejected. Unsafe `transferFrom` bypasses the hook entirely; such tokens are never accounted and have no rescue path `[RF-058]`.

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5. Ticket model

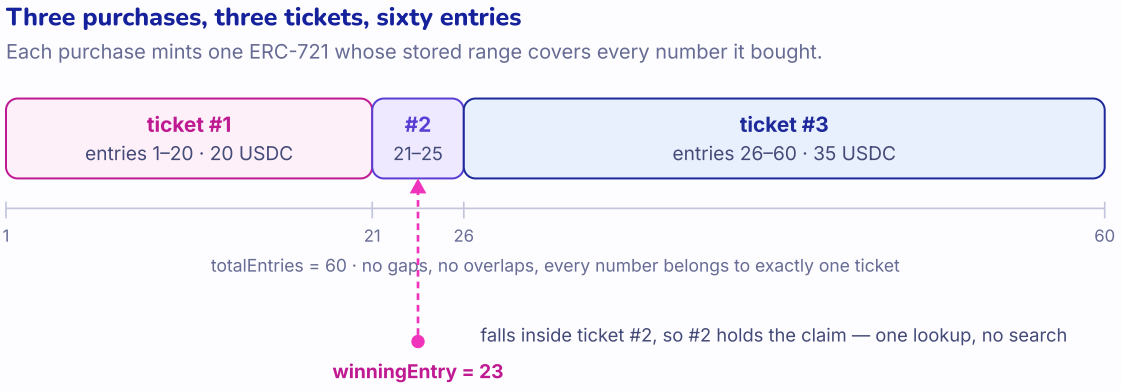
5.1 Issuance

`buyEntries(address recipient, uint128 entryCount) → uint256 ticketId`:

```
require status == Active                                    // InvalidStatus
require block.timestamp < endTime                           // SaleEnded
require recipient != 0                                      // InvalidRecipient
require entryCount != 0                                     // ZeroEntryCount
require entryCount <= type(uint128).max - totalEntries      // TotalEntriesOverflow

grossAmount = uint256(entryCount) * ENTRY_PRICE             // uint128 * 1e6 cannot overflow
before      = quoteToken.balanceOf(this)
quoteToken.safeTransferFrom(msg.sender, this, grossAmount)
received    = saturatingSub(quoteToken.balanceOf(this), before)
require received == grossAmount                             // UnsupportedQuoteToken

firstEntry  = totalEntries + 1
lastEntry   = totalEntries + entryCount
ticketId    = ticketCount + 1

totalEntries = lastEntry
ticketCount += 1
_ticketRanges[ticketId] = TicketRange(firstEntry, lastEntry)
unsettledPot += grossAmount

_safeMint(recipient, ticketId)
emit TicketPurchased(msg.sender, recipient, ticketId, firstEntry, lastEntry, entryCount, grossAmount)
```

The overflow guard precedes all token interaction `[RF-019]`. The balance-delta equality makes fee-on-transfer and rebasing tokens unusable by construction `[RF-016]`. `_safeMint` invokes `onERC721Received` on contract recipients; a revert propagates and rolls back the entire purchase including the payment `[RF-020]`. `nonReentrant` prevents nested purchases from a malicious receiver or a reentrant token.

Invariants. `grossSales() == totalEntries * ENTRY_PRICE` by construction, because `grossSales()` is a derived view with no backing storage `[RF-049]`. Ranges partition `[1, totalEntries]` exactly, with no gaps and no overlaps `[RF-018]`.



Ranges are assigned by advancing a single cursor, which is why they partition the sold entries exactly. Winner proof is a containment test against one stored range, independent of how many entries or tickets exist.

> **Integrator note.** `totalEntries` and `ticketCount` advance **independently**. A purchase of 20 entries advances the first by 20 and the second by 1. Code that conflates them is wrong. Both are `uint128` and uncapped; they must remain `bigint` end to end, because a JavaScript `number` conversion silently corrupts values above 2^53 `[RF-017]`.

5.2 Bearer semantics

Redemption and refund authorization is `msg.sender == ownerOf(ticketId)` — nothing else. Approvals and operator approvals are **insufficient** `[RF-022]`: an approval is a transfer right, not a claim right, so an operator wanting to redeem must first take the token. Both consuming paths `_burn` before any external asset transfer, so a right is consumed exactly once `[RF-023]`; a reverting transfer rolls the burn back and permits retry.

5.3 There is no transfer lock

```
lock ⟺ false
```

The `_update` override imposes exactly one restriction, and it is about *destination*, not *timing*:

```
if (to != address(0) && _isKnownProtocolDestination(to)) revert UnsafeProtocolDestination(to);
```

STATUS	NON-WINNING TICKET	WINNING TICKET
`Active`	transferable	n/a
`Drawing`	transferable	transferable
`NftWon`	transferable	transferable
`CashWon`	transferable	transferable
`Refunding`	transferable	transferable

Auditor note — this reverses the retired design. The previous generation froze all transfers during the draw and locked the winning ticket permanently after resolution. Both locks are gone `[RF-021]`. The reason is the settlement split (§9 and §6.5): because `settleWinningTicket` never reads `ownerOf`, a post-resolution transfer cannot corrupt accounting — it only changes who may later burn the ticket. The consequence is that a **known-winning** ticket is freely tradeable after settlement. That is intended behavior, not an oversight, and any marketplace integration must price it accordingly.

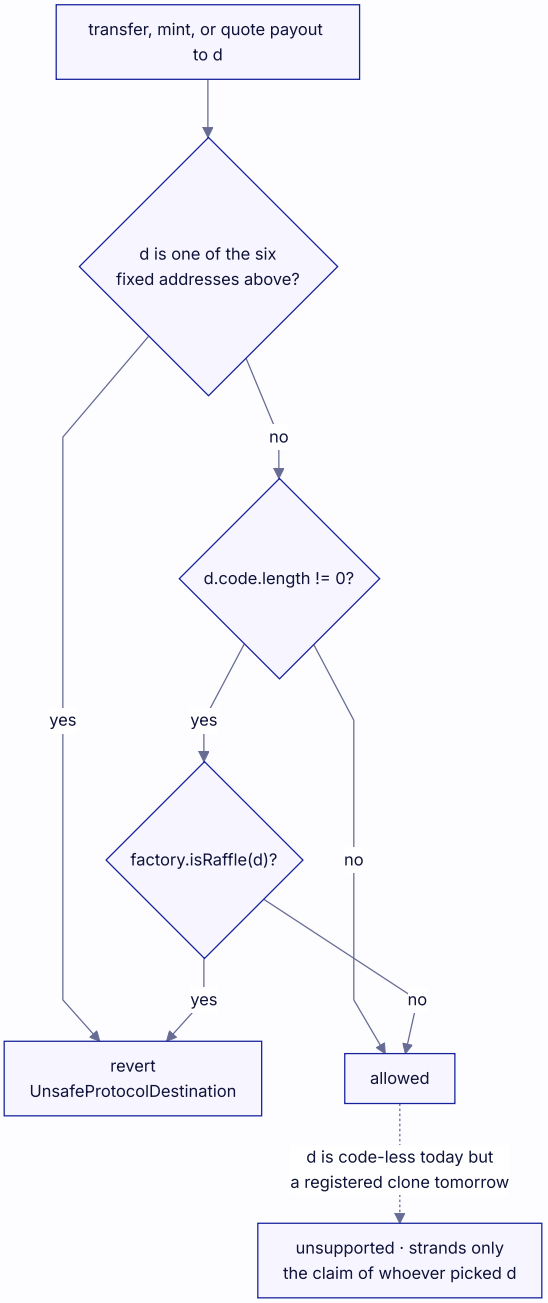
Integrator notes.

1. OpenZeppelin 5.x routes every transfer and both `safeTransferFrom` overloads through `_update`, so the destination screen is universal.
2. `_burn` also routes through `_update`, with `to == address(0)`; the guard's `to != address(0)` condition is what allows burns to succeed.
3. Losing tickets are **never** burned. After settlement they remain valid, transferable ERC-721s worth nothing `[RF-024]`. Do not present them as live claims.
4. `ticketRange()` is persistent historical metadata: it keeps returning a burned ticket's range. Live-ticket semantics must come from `ownerOf()` `[RF-018]`.

5.4 Destination screening

```
_isKnownProtocolDestination(d) =
      d == address(this)
   || d == factory
   || d == address(quoteToken)
   || d == address(i_vrfV2PlusWrapper)
   || d == address(prizeToken)
   || d == IRaffleFactory(factory).raffleImplementation()
   || (d.code.length != 0 && IRaffleFactory(factory).isRaffle(d))
```

Applied at `_update` (non-burn transfers) and `_transferQuoteExact`. An initialization-time analogue screens `sponsor`, `sponsorRecipient`, and `protocolTreasury`; the factory applies the same test to `prizeToken` and `sponsorRecipient` at creation, and to `protocolTreasury` in its own constructor `[RF-025]`, `[RF-003]`.



Destination screening, and the one gap it deliberately leaves open. The code-length test is what makes the residual reachable — and what keeps it confined to the party who chose the address.

> **Auditor note — the code-length residual.** The `isRaffle` branch fires only when the destination **already has code**. An address that is code-less today and later becomes a registered clone is *explicitly unsupported* `[RF-058]`. This is reachable — the repository tests it — but it is **owner-controlled self-stranding**: the holder must themselves choose the predicted address, and the regression is named `testFutureCanonicalCloneCanBrickOnlyItsOwnWinnerRedemption`. The retired design closed this gap with a permissionless recovery dispatcher; that dispatcher was itself exploitable — a permissionless `CREATE` caller could capture a predicted address and drain claims assigned before deployment — and was removed. The gap is accepted in preference to reintroducing a seizure-capable executor `[RF-072]`.

> **Auditor note — screening is same-factory only.** `isRaffle` resolves against *this* raffle's factory registry. A different factory's raffle is invisible to the check.

> **Auditor note — the treasury variant is a deployment trap, not a runtime one.** A factory deployed with a code-less address that later becomes a registered clone as its treasury can eventually halt creation at that nonce. The intended mainnet validator rejects a code-less treasury, making this a deployment control `[RF-071]`.

> **Gas note.** Every ticket transfer to an address with code performs an external `isRaffle` call to the factory. Integrators batching transfers should budget for it.

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6. Randomness integration

6.1 Request path

```
function requestDraw() external payable nonReentrant returns (uint256 requestId)
```

```
require status == Active                                    // InvalidStatus
require block.timestamp >= endTime                          // RaffleNotEnded
require totalEntries != 0                                   // NoEntriesSold
require block.timestamp < drawRequestDeadline()             // DrawRequestWindowExpired

quotedFee = i_vrfV2PlusWrapper.calculateRequestPriceNative(callbackGasLimit, 1)
require msg.value >= quotedFee                              // InsufficientVrfFee

status          = Drawing                   // written BEFORE the external call
drawRequestedAt = uint64(block.timestamp)
_requestInFlight = true
extraArgs        = ExtraArgsV1({ nativePayment: true })
(requestId, paidPrice) = requestRandomnessPayInNative(callbackGasLimit, requestConfirmations, 1, extraArgs)
require paidPrice <= msg.value                              // InsufficientVrfFee
vrfRequestId     = requestId
_requestInFlight = false

excess = msg.value - paidPrice
if excess != 0:
    raw call(gas(), msg.sender, excess, 0, 0, 0, 0)         // zero return-data copy
    require success                                         // NativeRefundFailed

emit DrawRequested(requestId, msg.sender, paidPrice, excess, drawRequestedAt, callbackDeadline())
```

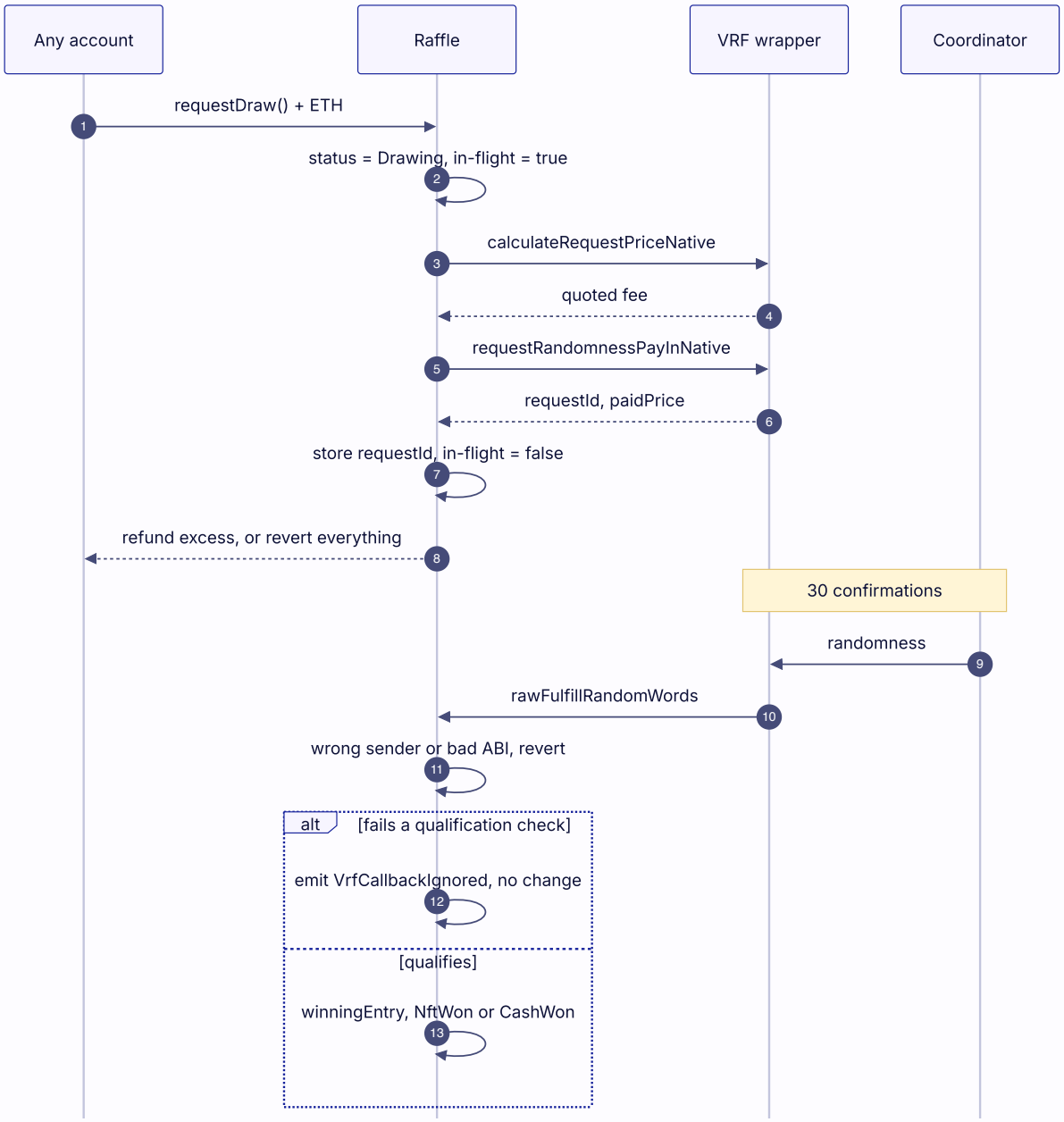
Key properties:

- **Exactly one request per raffle.** The `status = Drawing` write precedes the external call, so a second `requestDraw` fails the `Active` check, and a wrapper attempting to reenter `requestDraw` is rejected `[RF-030]`.
- **Quote and request use identical parameters.** Both pass `callbackGasLimit` and a word count of `1`, so a quote/request mismatch is structurally impossible `[RF-033]`.
- **The fee is paid from the caller's ETH, never the pot.** The USDC pot is untouched by the draw `[RF-030]`.
- **Excess is returned synchronously or the whole request reverts** `[RF-031]`. The raffle never holds a native balance: `receive()` reverts with `DirectNativeTransfer`.
- **Zero return-data copy.** The refund uses raw assembly with `retOffset = retSize = 0`, neutralizing a return-data bomb from a malicious requester.
- **Failure is not consuming.** A reverting fee read or request reverts the whole transaction, restoring `Active` with `drawRequestedAt == 0` and `vrfRequestId == 0`, preserving the request window for a retry `[RF-032]`.

6.2 Callback authentication

Authentication is layered, and each layer fails differently — which matters when reading logs:

1. **Transport (reverts).** The inherited `rawFulfillRandomWords` requires `msg.sender == i_vrfV2PlusWrapper`, the immutable wrapper. Anything else reverts.
2. **Decoding (reverts).** Calldata that cannot ABI-decode into `(uint256, uint256[])` reverts inside Solidity's decoder, before the handler body executes.
3. **Application (ignored, with an event).** `fulfillRandomWords` early-returns, emitting `VrfCallbackIgnored(receivedRequestId, expectedRequestId, status)`, when **any** of:
   - `_requestInFlight` — a synchronous callback arriving before `vrfRequestId` is stored;
   - `status != Drawing` — stale, duplicate, or post-refund delivery;
   - `requestId != vrfRequestId` — wrong request;
   - `randomWords.length != 1` — wrong word count;
   - `block.timestamp >= callbackDeadline()` — expired.



The full request and callback path. Note where the two failure styles diverge — transport and decoding revert, while every application-layer rejection returns quietly with an event so a rejected delivery has no provider-side consequence.

Auditor note. The handler **returns rather than reverts** on rejection `[RF-035]`. This is deliberate: reverting inside an oracle callback can have provider-side consequences. The trade-off is that a rejected callback is observable only as an event. Monitoring must alert on `VrfCallbackIgnored`; the subgraph indexes it as an immutable diagnostic entity with a per-raffle counter (`V1-SUBGRAPH-01`, closed).

Auditor note — the in-flight guard. `_requestInFlight` is the narrow defense for the window in which `vrfRequestId` is still zero `[RF-036]`. Without it, a wrapper that calls back synchronously — with request ID `0`, matching the not-yet-written slot — could resolve the raffle before the request even returned. Regressions cover the synchronous valid, wrong-ID, duplicate, and zero-ID attempts.

6.3 Resolution

```
resolvedEntry = uint128((randomWords[0] % uint256(totalEntries)) + 1)
winningEntry  = resolvedEntry
resolvedAt    = uint64(block.timestamp)
status        = totalEntries >= reserveEntries ? NftWon : CashWon
emit RaffleResolved(requestId, resolvedEntry, status)
```

That is the **entire** callback body after qualification. It performs bounded storage work and **zero external calls** — no ERC-20, no ERC-721, no user address, no loop, no ticket search `[RF-037]`. This is what makes a fixed 300,000-unit callback limit safe for a raffle with any number of entries or tickets, and it is asserted by a gas regression that holds both branches below the budget and shows callback cost does not scale with ticket count `[RF-063]`.

> **Integrator note.** `RaffleResolved` carries **no economic fields**. Unlike the retired design, the callback credits nothing: every liability is recorded later, by `settleWinningTicket` (§9). Index `WinningTicketSettled` for amounts, not `RaffleResolved`.

6.4 Modulo bias

`(random mod n) + 1` over a 256-bit domain is non-uniform whenever `n ∤ 2^256`. For `n ≤ 2^128` the total variation distance from uniform is bounded by roughly `n / 2^256`, so the absolute per-entry probability difference sits at the `2^-256` scale — negligible for any realistic entry count. It is **not zero**, and the protocol documents it rather than claiming uniformity `[RF-039]`, `[RF-075]`. No rejection sampling is performed; doing so would require either an unbounded callback loop or a second oracle round.

Consequence for public communication: raffle.fun must not be described as provably fair.

6.5 The oracle trust assumption

The randomness source is Chainlink VRF v2.5 through the official native direct-funding wrapper, using the exact-pinned official consumer base `[RF-034]`. Correctness of the drawn word rests on Chainlink's cryptographic and operational security model; the protocol adds no second source, no reroll, and no manual override `[RF-059]`.

Two properties are worth stating precisely for a reviewer:

Availability, not honesty, is the bounded failure. If the wrapper or coordinator is unavailable, misconfigured, or censored, the raffle does not hang: it falls to full, fee-free refunds at whichever deadline applies (§9). The contract cannot switch providers or raise its immutable 300,000-unit callback limit `[RF-059]`.

The request-time acquisition window is closed by ordering, not by a lock. Because the result is decided by a word delivered after `requestConfirmations = 30` confirmations, and because tickets are freely transferable, an observer who learns the word before it lands could in principle try to acquire the containing ticket. This is bounded by Chainlink's own model — the word is not revealed to the consumer before the callback — and by the fact that acquisition requires a willing counterparty who does not yet know the outcome either. It is **not** closed by a transfer lock, because there is none.

> **Auditor note.** A counterfeit *configured* wrapper would defeat all of this outright by choosing words. That is a deployment-validation concern, not a runtime one: the deployment validator pins and live-checks the official Chainlink wrapper, and the wrapper address is immutable thereafter (`V12` entry `243758`, `[RF-034]`).

---

7. Economic model

7.1 Constants

CONSTANT	VALUE	MEANING
`BPS`	10,000	basis-point denominator
`PROTOCOL_FEE_BPS`	500	5% protocol fee on gross
`CASH_WINNER_BPS`	8,000	80% winner share **of gross** in the cash branch
`ENTRY_PRICE`	1,000,000	one entry = 1 USDC at six decimals
`QUOTE_TOKEN_DECIMALS`	6	enforced by the factory constructor

All are compile-time constants embedded in the shared implementation. Neither rate can be changed for an existing raffle by any party, because no party exists `[RF-045]`, `[RF-002]`.

> **Auditor note — the split base changed.** The retired design computed the winner's 80% against a *post-fee* pot, yielding an effective 76/5/19. The current design takes all three shares from **gross**, yielding 80/5/15 `[RF-047]`. Any comparison against a historical report must account for this.

7.2 The sponsor's position

The reserve is not a safety rail bolted onto a raffle; it is the sponsor's **ask**, and because every entry is exactly one dollar, the reserve *is* the gross ask in whole USDC:

```
ask = reserveEntries × 1 USDC
```

Settlement is a binary on whether gross sales cleared that ask, and the sponsor is paid either way:

CONDITION	BRANCH	SPONSOR RECEIVES	PRIZE
`totalEntries >= reserveEntries`	`NftWon`	`0.95 × grossSales`, uncapped above the ask	to the winning ticket
`0 < totalEntries < reserve`	`CashWon`	`0.15 × grossSales` **and the NFT back**	returned to sponsor
`totalEntries == 0`	`Refunding`	nothing	returned to sponsor

If a sponsor wants a particular **net** amount, the gross reserve is `reserveEntries = ceil(ask / 0.95)`.

This is structurally a **covered call**. The sponsor writes an option on an asset they hold; entry buyers pay the premium; if the strike is cleared the asset is delivered, and if it is not, the writer keeps both the asset and a share of the premium. `reserveEntries` is the strike, one dollar per entry is the premium unit, and `endTime` is expiry. Three differences from a conventional covered call matter:

1. **Exercise is probabilistic per buyer but deterministic in aggregate.** Clearing the ask guarantees the NFT transfers; which buyer receives it is the random part.
2. **The writer cannot close the position early.** The prize is escrowed until settlement `[RF-013]`, so there is no buy-back and no cancellation.
3. **The unexercised premium is shared, not kept whole.** In the cash branch the sponsor keeps 15% of gross, not the full premium, because 80% is paid to the drawn ticket.

7.3 Fee

```
protocolFee = floor(grossPot × 500 / 10_000)
```

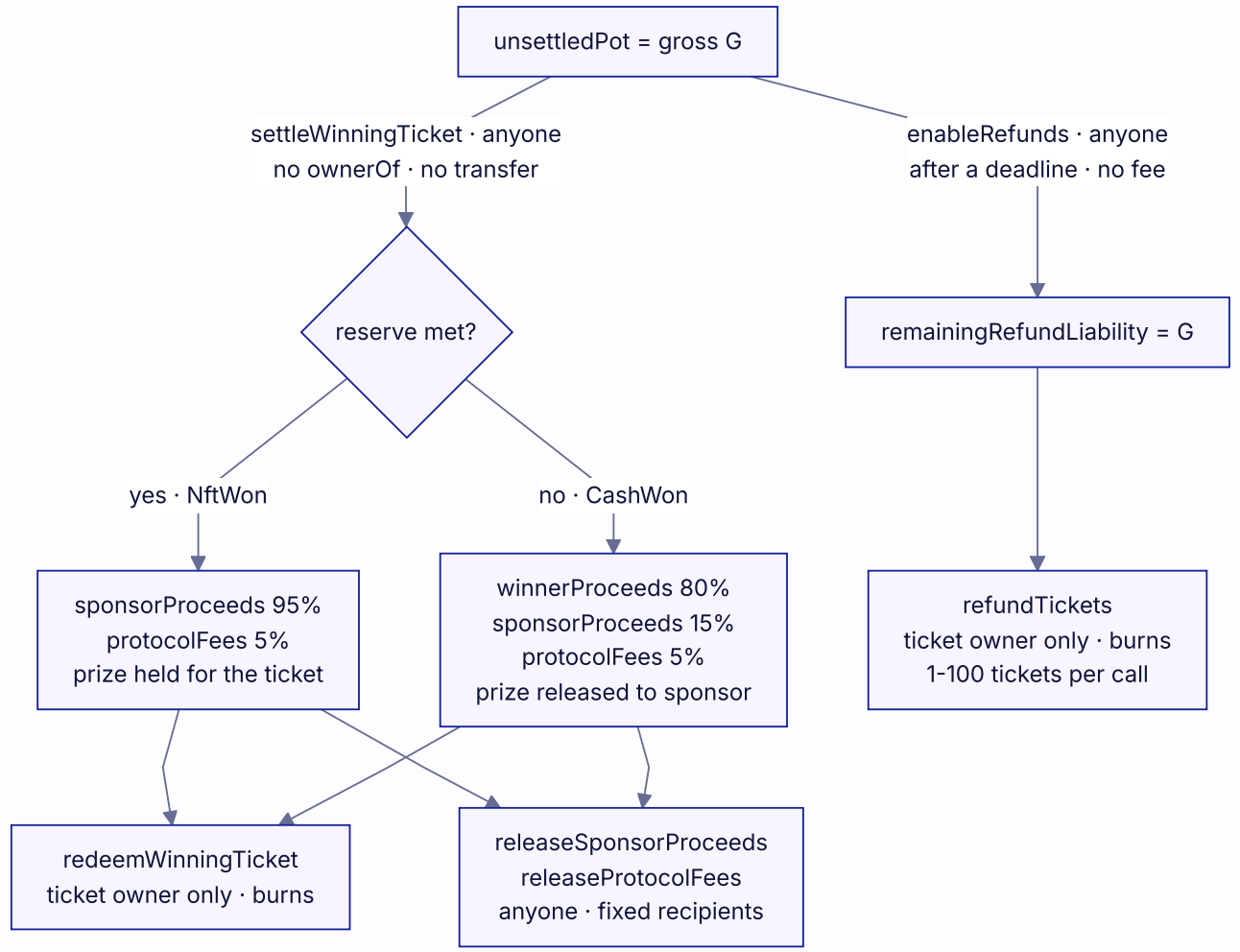
Computed via `Math.mulDiv` in exactly **one** location — `_settleWinningTicket` — over the whole pot rather than per purchase, which avoids per-purchase rounding accumulation. Charged on neither refund origin nor on an empty raffle `[RF-045]`, `[RF-053]`.

7.4 Branch settlement

Settlement is a single private routine reached from either `settleWinningTicket` or `redeemWinningTicket` `[RF-041]`:

```
require !settlementComplete                                  // SettlementAlreadyComplete
require status ∈ {NftWon, CashWon}                           // InvalidStatus
require ticketRange(ticketId) contains winningEntry          // TicketDoesNotContainWinningEntry

grossPot    = unsettledPot
protocolFee = mulDiv(grossPot, 500, 10_000)

if status == NftWon:
    sponsorAmount = grossPot − protocolFee                   //  95%
else:
    cashAmount    = mulDiv(grossPot, 8_000, 10_000)          //  80%
    sponsorAmount = grossPot − protocolFee − cashAmount      //  15%
    winnerProceeds = cashAmount

settlementComplete = true
winningTicketId    = ticketId
unsettledPot       = 0
sponsorProceeds    = sponsorAmount
protocolFees       = protocolFee

emit WinningTicketSettled(ticketId, msg.sender, status, cashAmount, protocolFee, sponsorAmount)
```

Note what is absent: no `ownerOf`, no `_burn`, no token transfer, no user call. Settlement is pure accounting `[RF-041]`.



Where the pot goes, and who may move it. Recording a liability is permissionless and ownership-blind; paying it out is either owner-gated or fixed-recipient, and no caller ever names a destination.

Refund branches. `remainingRefundLiability = unsettledPot`, `unsettledPot = 0`; each burned ticket pays exactly `(lastEntry − firstEntry + 1) × ENTRY_PRICE` `[RF-052]`.

7.5 Conservation

Both floor divisions assign their remainder to the sponsor via subtraction, so value is conserved exactly in raw token units `[RF-048]`:

```
NftWon    : protocolFee + sponsorProceeds                 = grossPot
CashWon   : protocolFee + winnerProceeds + sponsorProceeds  = grossPot
Refunding : Σ (range weight × ENTRY_PRICE per burned ticket) = grossPot
```

Worked example (adversarial rounding). The subtraction rule is what makes conservation hold even for a pot that is not a clean multiple. For `grossPot = 999,999` raw units:

QUANTITY	COMPUTATION	VALUE
`protocolFee`	`floor(999,999 × 500 / 10,000)`	49,999
`winnerProceeds`	`floor(999,999 × 8,000 / 10,000)`	799,999
`sponsorProceeds`	`999,999 − 49,999 − 799,999`	150,001
Sum		**999,999** ✓

> **Auditor note.** In practice `grossPot` is always a multiple of `1,000,000`, because entries are whole dollars and the entry price is fixed `[RF-014]`. Both floors are therefore exact and the remainder is zero. The example above is unreachable through `buyEntries`; it is included to show the accounting is correct regardless, which is what a fuzz campaign over the arithmetic actually exercises.

7.6 Canonical cash-branch example

80 entries against a 100-entry reserve, gross 80.00 USDC:

RECIPIENT	AMOUNT
Winning ticket (current owner)	64.00 USDC
Protocol treasury	4.00 USDC
Sponsor recipient	12.00 USDC
Sponsor recipient	the NFT

7.7 Canonical NFT-branch example

100 entries against a 100-entry reserve — equality meets the reserve `[RF-038]` — gross 100.00 USDC:

RECIPIENT	AMOUNT
Winning ticket (current owner)	the NFT
Protocol treasury	5.00 USDC
Sponsor recipient	95.00 USDC

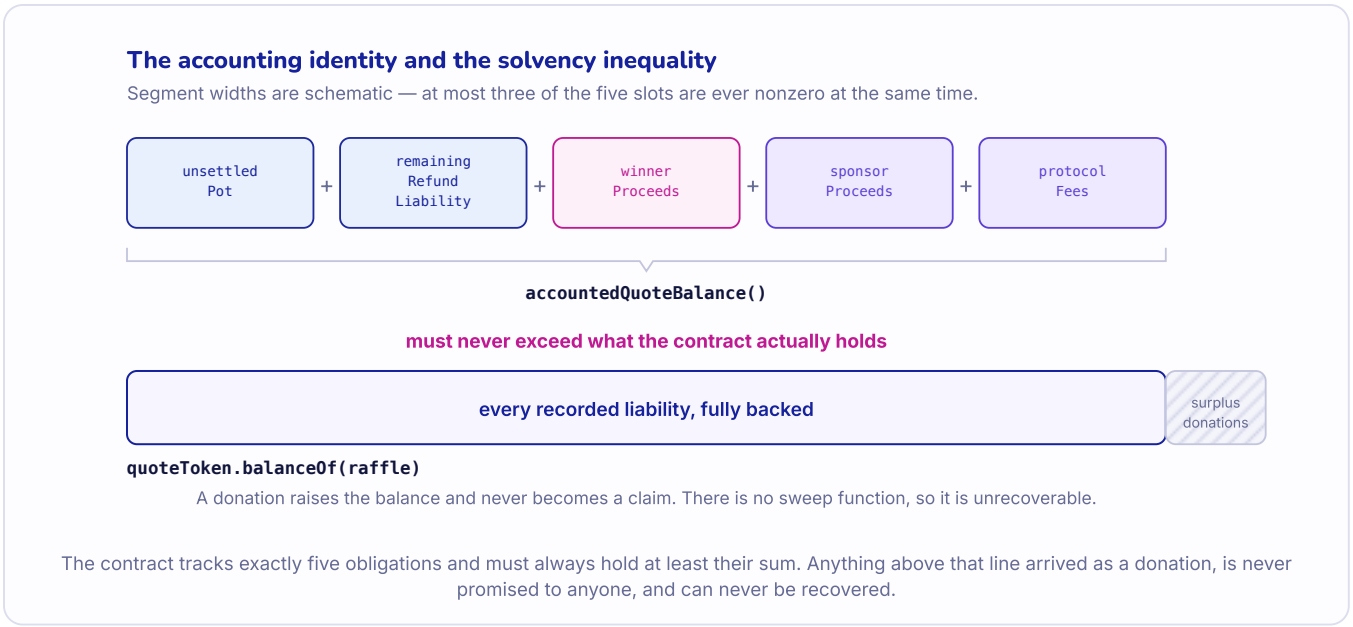
8. Accounting and solvency

8.1 The identity

```
accountedQuoteBalance()
  = unsettledPot
  + remainingRefundLiability
  + winnerProceeds
  + sponsorProceeds
  + protocolFees
```

8.2 Solvency invariant

```
quoteToken.balanceOf(raffle) >= accountedQuoteBalance()
```



Surplus arises only from direct donations and never becomes a liability `[RF-054]`. There is no sweep function; donated quote tokens are unrecoverable, a deliberate consequence of G1.

8.3 Liability lifecycle by status

STATUS	UNSETTLEDPOT	REMAININGREFUNDLIABILITY	WINNERPROCEEDS	SPONSORPROCEEDS	PROTOCOLFEES
`Active`	grows with sales	0	0	0	0
`Drawing`	frozen at gross	0	0	0	0
`NftWon` (unsettled)	**full gross**	0	0	0	0
`NftWon` (settled)	0	0	0	95%	5%
`CashWon` (unsettled)	**full gross**	0	0	0	0
`CashWon` (settled)	0	0	80%	15%	5%
`Refunding`	0	full gross (decreasing)	0	0	0

> **Auditor note.** Both resolved branches sit at "full gross in `unsettledPot`" until somebody settles. Settlement is permissionless and free of external interaction specifically so that this state is trivially exitable by any party `[RF-041]`.

8.4 Exact-transfer verification

`_transferQuoteExact(to, amount)` verifies **both sides** `[RF-055]`:

```
require !_isKnownProtocolDestination(to)                  // InvalidQuoteDestination
raffleBefore = balanceOf(this);  recipientBefore = balanceOf(to)
safeTransfer(to, amount)
debited  = saturatingSub(raffleBefore, balanceOf(this))
credited = saturatingSub(balanceOf(to), recipientBefore)
require debited == amount && credited == amount           // UnsupportedQuoteTokenTransfer
```

Checking both sides catches recipient-bonus tokens (credit > debit) and sender-rebate tokens (debit < amount) that a one-sided check would miss. On revert, the ticket burn, the zeroed liability slot, and the decremented refund liability are all restored, so the claim survives for a later retry. This preserves the onchain claim across a USDC blocklist event — it cannot force the transfer `[RF-069]`.

Incoming payment is verified symmetrically at `buyEntries` `[RF-016]`. Settlement performs **no** external asset interaction at all, so there is nothing to verify there.

---

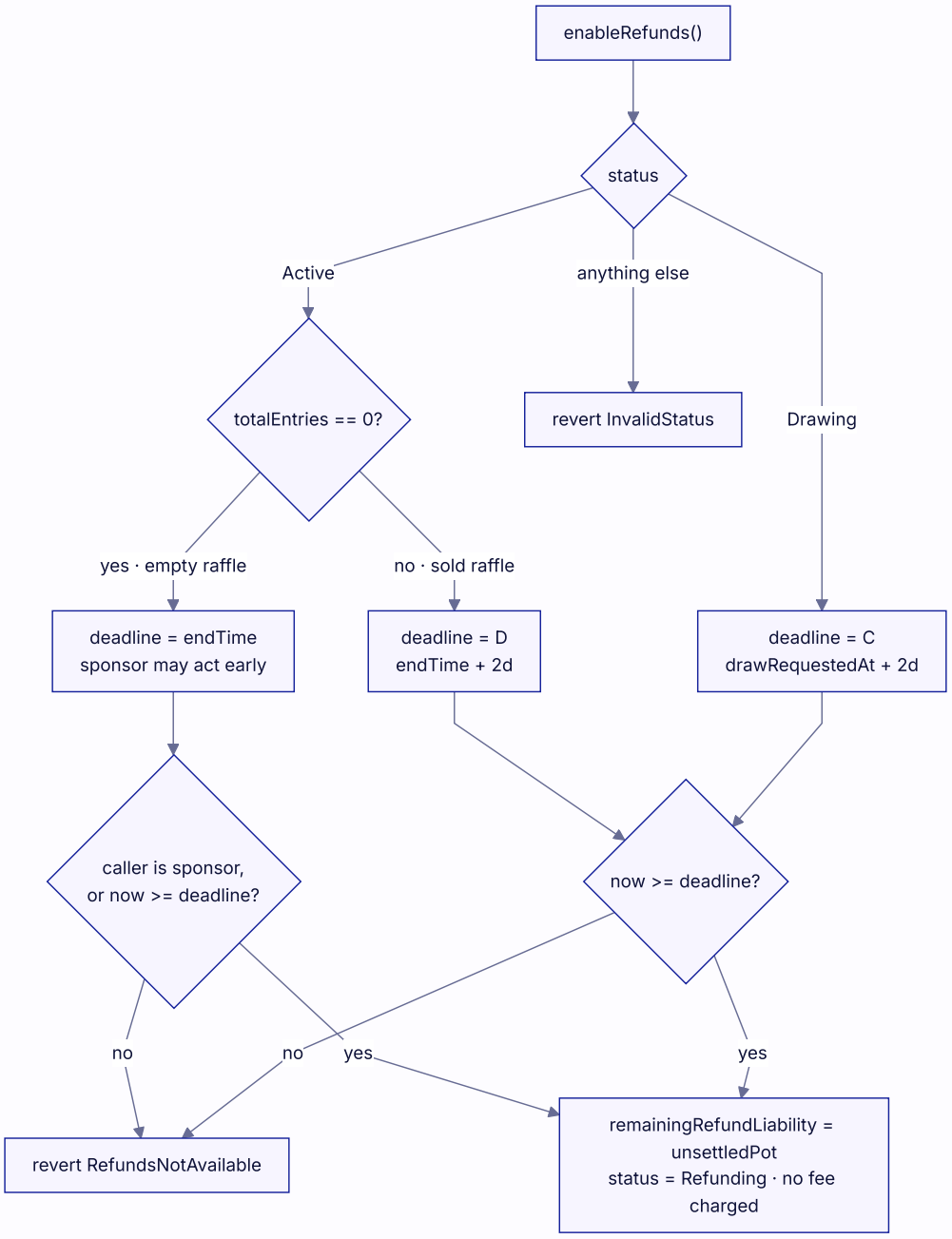
9. Liveness analysis

9.1 The refund origins

`enableRefunds()` is permissionless and dispatches on status `[RF-050]`:

#	FROM	DEADLINE	WHO MAY CALL	INTERPRETATION
1	`Active`, `totalEntries == 0`	`endTime`	sponsor any time; anyone at/after	Empty raffle, zero liability
2	`Active`, `totalEntries != 0`	`drawRequestDeadline()`	anyone at/after	No randomness request was accepted
3	`Drawing`	`callbackDeadline()`	anyone at/after	Request accepted, no valid callback

Any other status reverts `InvalidStatus`. Before the deadline: `RefundsNotAvailable`.



One permissionless function covers all three stalls. Every path either reverts or lands on the same terminal state, and none of them charges a fee.

Auditor note. Origin 1 is the **only** sponsor-specific timing privilege anywhere in the protocol, and it can be exercised only when there is no buyer money at stake `[RF-051]`. It replaces the retired design's separate `Closed` state; an empty raffle simply becomes zero-liability `Refunding`, from which `releaseSponsorPrize` returns the NFT.

9.2 Hard cutoffs, not races

A deadline does not mutate state. It changes which transactions are valid. The design makes success and refund *mutually exclusive by construction* rather than by inclusion order `[RF-028]`, `[RF-029]`:

- `requestDraw` requires `now < D`; refund origin 2 requires `now >= D`.
- A callback resolves only when `now < C`; refund origin 3 requires `now >= C`.

So at the exact boundary instant, the "success" transaction is already invalid and only the refund transition is available. A late callback is not merely losing a race — it is ignored outright, emitting `VrfCallbackIgnored`, **even if `enableRefunds` has not yet been called** `[RF-035]`. Regressions assert this at equality in both orderings.

> **Auditor note.** This differs materially from the retired design, where a late callback could still resolve the raffle if nobody had finalized refunds first. The current semantics remove that ambiguity, at the cost of turning any post-cutoff censorship or reorganization into a forced refund rather than a delayed success `[RF-070]`.

9.3 Refund redemption

```
function refundTickets(uint256[] calldata ticketIds) external returns (uint256 amount)
```

Requires `status == Refunding` and `1 <= ticketIds.length <= 100`. Each ID must be owned by `msg.sender`; each is burned and its range weight accumulated. Then `amount = aggregateEntries × ENTRY_PRICE`, `remainingRefundLiability -= amount`, and `_transferQuoteExact(msg.sender, amount)` `[RF-052]`.

The batch is **atomic**: a duplicate ID fails on the second `ownerOf` (the first burn cleared the owner), and a foreign ID fails ownership — either reverts the whole batch.

> **Integrator note — the bound is on tickets, not entries.** One hundred is the maximum number of **ticket IDs** per call. A single ticket may carry any `uint128` entry count, so one call can refund an arbitrarily large amount of USDC. Refund gas scales with the batch length, not with the entries inside the tickets `[RF-063]`.

9.4 Redemption liveness

The winner never depends on a third party. `redeemWinningTicket` performs settlement inline when settlement has not yet occurred, so the owner can settle and redeem in one transaction `[RF-042]`. Conversely, if settlement was already committed by someone else, the winner's claim is unaffected and the sponsor and treasury claims are independently releasable `[RF-044]`, so winner inactivity or a broken prize cannot block those.

9.5 What liveness does not guarantee

Three limits, stated plainly:

1. **Deadlines are deterministic only given inclusion.** They cannot defeat validator or builder censorship, a halted chain, or a reorganization. Censorship or a reorganization that removes an otherwise valid request or callback *after* its cutoff prevents replay and forces the refund branch `[RF-070]`.
2. **Nobody is obliged to request the draw.** The requester pays Chainlink's fee from their own funds with no protocol reimbursement. The economic assumption is that a ticket holder or the sponsor is motivated to pay it; if nobody does, the raffle refunds in full `[RF-030]`.
3. **A resolved result is final, including when the prize cannot be delivered.** There is no post-result refund timeout in either branch `[RF-040]`. If the prize collection is later paused, upgraded, frozen, burned, or hostile, the NFT winner's claim can be blocked indefinitely and buyers are **not** refunded — while the sponsor and treasury quote claims remain releasable. This is the sharpest consequence of the bearer redesign and the single most material launch consideration `[RF-068]`; §12 T3 and §16 restate it.

---

10. Authority model

10.1 Complete authority matrix

CAPABILITY	SPONSOR	SPONSOR RECIPIENT	TICKET OWNER	TREASURY	ANYONE
Create a raffle	✓	—	—	—	✓
Change any existing raffle parameter	×	×	×	×	×
Cancel or pause a **sold** raffle	×	×	×	×	×
Pause or reconfigure the factory	×	×	×	×	×
Close a zero-sale raffle before `endTime`	✓	×	×	×	×
Close a zero-sale raffle at/after `endTime`	✓	✓	✓	✓	✓
Request the draw	✓	✓	✓	✓	✓
Choose or override the winner	×	×	×	×	×
Enable refunds after a deadline	✓	✓	✓	✓	✓
Settle the winning ticket	✓	✓	✓	✓	✓
Redirect settlement value	×	×	×	×	×
Redeem the winning ticket	—	—	✓ (owner only)	×	×
Refund tickets	—	—	✓ (owner only)	×	×
Trigger release of sponsor proceeds	✓	✓	✓	✓	✓
Trigger release of protocol fees	✓	✓	✓	✓	✓
Trigger return of the prize (Cash/Refunding)	✓	✓	✓	✓	✓
Rescue arbitrary assets	×	×	×	×	×

Read the "Anyone" column together with the "Redirect" row: many actions are permissionless to *trigger*, and none of them let the caller choose a destination `[RF-044]`.

10.2 There is no factory owner

`RaffleFactory` inherits only `IRaffleFactory` and `ReentrancyGuard`. It declares no `Ownable`, no role, no `onlyOwner` modifier, no setter, and no pause flag. Its entire external mutating surface is `createRaffle` `[RF-001]`, `[RF-004]`.

Everything a factory owner could once reach is now fixed at factory deployment: the quote token, the VRF wrapper, the protocol treasury, the raffle implementation, the callback gas limit, and the confirmation count are all `immutable` or `constant` `[RF-002]`. Changing any of them requires deploying a new factory; existing raffles are unaffected by, and invisible to, a newer factory.

> **Auditor note.** This removes the retired design's owner-key trust assumption, its treasury setter, its creation pause, and the renunciation-bricking hazard, all at once. The corresponding cost is in §10.3 and is not small.

10.3 Incident response envelope

Existing raffles cannot be upgraded, paused, or patched, and **new creation cannot be stopped onchain** `[RF-009]`, `[RF-074]`. A confirmed vulnerability permits only: warn users, disable first-party frontend writes and

sponsor onboarding, monitor, and deploy a new factory version. Raffles already running proceed to their own terminal states regardless.

11. Failure mode analysis

FAILURE	DETECTION	OUTCOME	FEE CHARGED	FACT
Prize transfer or activation fails at creation	double post-condition	entire creation reverts, clone discarded	—	RF-010
Prize reenters during escrow	`nonReentrant`	creation reverts atomically	—	RF-010
Quote token under- or over-delivers on purchase	balance delta	purchase reverts, no ticket	—	RF-016
Ticket receiver reverts	`_safeMint`	purchase fully reverts, payment returned	—	RF-020
Cumulative entry overflow	pre-transfer guard	reverts before payment	—	RF-019
Zero entries sold	`totalEntries == 0`	zero-liability `Refunding`; NFT returned	**no**	RF-051
VRF fee read or request reverts	no `try/catch`	tx reverts, stays `Active`, window preserved	—	RF-032
Native excess refund fails	raw call result	whole request reverts	—	RF-031
Direct ETH sent to raffle	`receive()`	reverts	—	RF-031
No draw request within 2 days of sale end	`enableRefunds` origin 2	full refunds	**no**	RF-050
No valid callback within 2 days of the request	`enableRefunds` origin 3	full refunds	**no**	RF-050
Synchronous / wrong-ID / stale / duplicate callback	five application guards	ignored, `VrfCallbackIgnored` emitted	—	RF-035
Wrong word count	`randomWords.length != 1`	ignored, event emitted	—	RF-035
Callback at or after `callbackDeadline()`	timestamp guard	ignored, refunds available	—	RF-035
Unauthorized or undecodable callback calldata	transport / ABI decoder	reverts	—	RF-035
Non-winning ticket supplied to settlement	range containment	reverts `TicketDoesNotContainWinningEntry`	—	RF-041
Second settlement attempt	`settlementComplete`	reverts	—	RF-041
Non-owner attempts redemption	`ownerOf` check	reverts `NotTicketOwner`	—	RF-022

FAILURE	DETECTION	OUTCOME	FEE CHARGED	FACT
Prize undeliverable at redemption	`ownerOf` post-check	tx reverts; burn rolled back; retry possible	—	`RF-043`
Prize permanently undeliverable	none	**winner claim blocked; buyers are NOT refunded**	yes*	`RF-068`
Quote payout under/over-delivers	two-sided delta	reverts; claim preserved	—	`RF-055`
Payout or ticket destination is a protocol sink	`_isKnownProtocolDestination`	reverts; claim preserved	—	`RF-025`
Forced ETH (`SELFDESTRUCT`)	none	outside accounting; unrecoverable	—	`RF-058`
Direct quote-token donation	none	surplus; unrecoverable	—	`RF-054`
Unrelated NFT via unsafe transfer	none	outside accounting; unrecoverable	—	`RF-058`
Ticket sent to an incapable contract	none	claim forfeit; **no recovery**	—	`RF-072`
Claim assigned to a future code-less clone address	none	**unsupported**; strands only that party's own claim	—	`RF-058`
Losing tickets after settlement	none	remain valid, tradable, worth nothing	—	`RF-024`

* The fee and sponsor proceeds are recorded at settlement and remain releasable even when prize delivery is permanently blocked. That asymmetry is deliberate — one broken recipient must not roll back another's allocation — and it is exactly why prize-collection review is a product control (§12 T3).

12. Trust assumptions

Ordered by the project's own severity assessment.

T1 — Chainlink VRF v2.5 wrapper and coordinator

The configured wrapper and coordinator must be the official deployments and must remain available, correctly priced, and within the coordinator's maximum gas. An outage, configuration change, coordinator failure, or prolonged censorship yields refunds rather than a wrong result `[RF-059]`. A *counterfeit configured* wrapper would be able to choose words outright — hence deployment pinning is a mandatory gate, not a convenience `[RF-034]`.

Deployment validation must confirm the 300,000-unit consumer limit plus wrapper overhead plus Chainlink's EIP-150 compensation, `floor(callbackGasLimit / 63) + 1`, fits the live coordinator maximum, at the finalized release block (`V1-DEPLOY-01`, closed) `[RF-033]`.

T2 — Quote-token issuer

The intended quote token is USDC. Circle can pause, freeze, blocklist, or upgrade it. Exact-delta checks preserve onchain claims across a failed transfer but cannot force one `[RF-069]`. The Solidity is **not** hard-coded to USDC — the factory accepts any code-bearing address reporting six decimals, and "official USDC" is enforced by deployment validation and human review `[RF-057]`.

T3 — Prize ERC-721 honesty (most material)

ERC-165 support is self-reported, and `ownerOf` verification routes through the same contract that could be lying. A malicious, upgradeable, pausable, transfer-restricted, or later-hostile collection can forge custody statements at creation or permanently block delivery after a final result, and no contract can assess whether a prize has value `[RF-056]`, `[RF-068]`.

Because §9.5(3) removes the post-result refund fallback, this dependency is the most material launch consideration in the current risk set. Prize admission and collection review are product controls, and per-raffle collection verification belongs at the purchase decision, not only in general documentation.

T4 — Ethereum inclusion, ordering, and finality

Ordering at deadline boundaries, request/callback delay, individual censorship, halt, and reorganization are all outside protocol control. Thirty confirmations materially reduce ordinary reorganization risk without creating a mathematical finality guarantee `[RF-070]`.

T5 — Deployment configuration and the treasury wallet

The factory's immutables cannot be corrected after deployment. The treasury must be an independently reviewed contract wallet with tested signer, recovery, module, and monitoring policies; a code-less or predicted-clone treasury can poison a factory `[RF-071]`, `[RF-079]`. Verified source, runtime hashes, the canonical clone target, and official dependency addresses must all be checked against a finalized release-day block.

T6 — User destination choices

Arbitrary non-callable ticket, proceeds, and prize destinations are deliberately unsolved because they are undecidable from bytecode `[RF-072]`. Every reachable case is self-stranding: the party choosing the destination is the only party harmed.

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13. Verification evidence

> These are the maintainers' own campaign results. They are evidence of testing depth, not proof of correctness, and **not an audit** `[RF-066]`.

13.1 Freshness caveat (read first)

The totals in §13.2 were captured at implementation SHA `92eccb4` with mutation evidence at `e9e0e73`. They predate the hard request/callback-boundary remediation, the official Chainlink consumer-base migration, the bearer-redemption redesign, and the ownerless-factory change. They are preserved evidence for those SHAs and **do not** validate the current source `[RF-067]`.

The deterministic suites were reproduced at **this document's commit** and are the only totals here that describe the current source:

SUITE	RESULT AT `E65E1E5`
Foundry (81 tests)	**80 passed, 0 failed, 1 skipped** (RPC-gated fork)
Hardhat deployment + journey	**21 passed, 0 failed**

This matches the independent run recorded in `V12-REVIEW-2026-08-20.md` at SHA `3da958f`. Everything in §13.2 other than these two rows still requires reproduction from a clean checkout of the frozen release SHA (`V1-REL-02`).

13.2 Recorded campaign totals

GATE	RESULT
Foundry deterministic/security/fuzz/invariant	72 passed, 0 failed
RPC-gated Ethereum fork test	1 **skipped** without an endpoint
Hardhat deployment and journey suite	22 passed, 0 failed
Independent Python protocol model	11 passed, 0 failed
Declared mutation campaign	52 of 52 mutants killed
Deterministic gas suite	57 passed, 0 failed
SDK / web / subgraph	14 / 15 / 7 passed, 0 failed
Production-only coverage	100.00% lines, 100.00% functions, 94.12% branches
Slither	47 contracts, 64 detectors, 0 results
Gitleaks	tracked worktree and 25-commit history scans: 0 leaks

Eight Foundry fuzz properties passed the 1,000-case default campaign and the 100,000-case audit profile. Seven stateful invariants passed the default 16,384-call/property profile and the 256,000-call/property audit and strict profiles; the strict profile enables `fail_on_revert` and completed with zero handler reverts.

13.3 Invariant catalog

The "seven stateful invariants" above is the count at SHA `92eccb4`. The current source ships **nine**, the two additions covering the hard deadline boundaries and the settlement/redemption markers introduced by the remediation:

INVARIANT	PROPERTY
`invariantTicketIdsAreSequential`	ticket IDs are dense and ordered
`invariantRangesPartitionEverySoldEntryExactlyOnce`	ranges partition `[1, totalEntries]`
`invariantWinningEntryHasExactlyOneReceiptProof`	exactly one ticket can satisfy winner proof
`invariantQuoteAccountingIsExactAndSolvent`	§8.1 identity and §8.2 solvency hold
`invariantBranchEconomicsRemainEightyFifteenFiveNinetyFiveFiveOrFullRefund`	§7.5 conservation in all three branches
`invariantStatusAndTerminalTransitionsAreMonotonic`	terminal statuses are absorbing
`invariantDrawAndCallbackDeadlinesAreHardAndOrdered`	§9.2 hard cutoffs
`invariantSettlementAndRedemptionMarkersMatchTicketState`	markers agree with the ERC-721 ledger
`invariantPrizeCustodyMatchesClaimMarker`	`prizeClaimed` agrees with actual custody

`docs/SECURITY-INVARIANTS.md` enumerates 64 review-level invariants mapped to unit, adversarial, fuzz, stateful, Echidna, fork, and static evidence.

13.4 Fork validation

Two RPC-gated cases exist — `testEthereumMainnetChainlinkVrfUsdcAndRangeReceipt` and `testEthereumSepoliaChainlinkVrfUsdcAndRangeReceipt` — reading live USDC and Chainlink wrapper state. **Both**

were compiled but skipped in the recorded runs; no RPC-backed result is claimed for this candidate `[RF-034]`. **No transaction has been broadcast on any network** `[RF-065]`.

13.5 Acknowledged tooling limits

- The Echidna harness compiles, but **no current runtime campaign is claimed** because the executable was unavailable (`V1-REL-05`).
- The 52-mutant result covers a declared, hand-selected compiling set. It is **not** an exhaustive mutation-space claim.
- Fuzzing, stateful invariants, differential modelling, and coverage cannot enumerate all states or external behaviors.
- Static analysis, dependency, signature, secrets, ABI-drift, gas, size, coverage, source verification, and deployment validation must all be rerun on the frozen release SHA (`V1-REL-02`, `V1-REL-03`).

13.6 Independent review of a third-party export

`V12-REVIEW-2026-08-20.md` reviewed a fifteen-entry third-party finding export against SHA `3da958f` and promoted **none** of it to a confirmed in-scope exploit. Several entries remain useful as residual-risk evidence and are folded into §11 and §12:

ENTRY	SUBJECT	DISPOSITION HERE
`243747`	Untrusted prize tokens can forge escrow/delivery	§12 T3, `RF-056`
`243754`	Unavailable prize can lock winner claims	§9.5(3), §12 T3, `RF-068`
`243748` / `243749` / `243759`	Future-clone claim trapping	§5.4, `RF-072`
`243750`	Future-clone treasury poisons a factory	§12 T5, `RF-071`
`243751` / `243752`	`uint64` timestamp wrap	Invalid on Ethereum, `RF-027`
`243757` / `243758`	Quote/wrapper admission breadth	§12 T1, T2, deployment gates
`243760`	`ticketRange()` lacks live-existence semantics	§5.3 integrator note

The review's own follow-up list — direct refund regressions for duplicate/nonexistent IDs and an exactly-100 batch, a fully lying ERC-721 mock, a predicted-treasury regression, and per-raffle collection warnings at the purchase decision — remains open hardening work.

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14. Integration guide

14.1 Authenticating a raffle

Never trust a contract because it looks like a raffle. There is no Lens to do this for you. Authenticate through the registry `[RF-005]`:

```
require(factory.isRaffle(candidate), "not canonical");
```

Bytecode shape is not proof: an ERC-1167 clone of the same implementation deployed outside the factory is unregistered and must be rejected. Conversely `isRaffle` proves canonical deployment only — never the quality or legitimacy of the prize collection `[RF-004]`.

14.2 Reading state

Read the raffle directly. The relevant views:

GROUP	VIEWS
Lifecycle	`status()`, `endTime()`, `drawRequestDeadline()`, `callbackDeadline()`, `drawRequestedAt()`, `resolvedAt()`
Sale	`totalEntries()`, `ticketCount()`, `grossSales()`, `reserveEntries()`, `ENTRY_PRICE()`
Tickets	`ticketRange(id)`, `ownerOf(id)`, `winningEntry()`, `winningTicketId()`
Liabilities	`unsettledPot()`, `remainingRefundLiability()`, `winnerProceeds()`, `sponsorProceeds()`, `protocolFees()`, `accountedQuoteBalance()`
Markers	`settlementComplete()`, `winnerRedeemed()`, `prizeClaimed()`, `winnerRecipient()`
Fixed	`factory()`, `quoteToken()`, `vrfWrapper()`, `prizeToken()`, `prizeTokenId()`, `sponsor()`, `sponsorRecipient()`, `protocolTreasury()`, `raffleId()`
VRF pricing	`getVrfRequestPrice()`, `estimateVrfRequestPrice(gasPriceWei)`

Caveats:

- `callbackDeadline()` returns **0** before any request. Do not render it as an epoch.
- `getVrfRequestPrice()` is a live wrapper quote and moves with gas pricing. Quote immediately before sending, and overpay to absorb drift — excess is returned `[RF-031]`. `estimateVrfRequestPrice` exists for what-if pricing at a hypothetical gas price.
- `grossSales()` counts everything ever sold; it is **not** the remaining pot `[RF-049]`.

14.3 Write paths

All first-party SDK actions simulate against live chain state before writing. The subgraph is **never** authoritative for transactions.

ACTION	NOTES
`createRaffle`	requires prior ERC-721 approval; sale starts immediately on success
`buyEntries`	requires prior ERC-20 approval for `entryCount × 1 USDC`; any positive `uint128` count
`requestDraw`	**payable**; quote immediately before, overpay to absorb drift; valid only in `[endTime, D)`
`enableRefunds`	permissionless; check the applicable deadline first
`settleWinningTicket`	permissionless; supply the ticket ID whose range contains `winningEntry`
`redeemWinningTicket`	caller must be `ownerOf(ticketId)`; settles inline if needed
`refundTickets`	batch of 1–100 **ticket IDs**, deduplicated, all owned by the caller
`releaseSponsorProceeds` / `releaseProtocolFees` / `releaseSponsorPrize`	permissionless; always pay the fixed recipient

14.4 Indexing pitfalls

1. `RaffleResolved` carries **no amounts**. Index `WinningTicketSettled` for the fee, sponsor, and winner allocations (§6.3).
2. `winningTicketId` is `0` until settlement, even though `winningEntry` is set at resolution. Key "resolved" on `resolvedAt` / `status`, and "settled" on `settlementComplete`.
3. Tickets are transferable in **every** status, including after settlement. An indexer must track ERC-721 `Transfer` events throughout the lifecycle and must not assume the settler or the resolution-time owner is the eventual

redeemer `[RF-021]`.

4. Losing tickets are never burned and remain transferable forever `[RF-024]`. `ticketRange()` also survives a burn, so use `ownerOf()` for live-ticket semantics.
5. `VrfCallbackIgnored` is emitted **without** a state change — a monitoring signal, not a lifecycle event. It is indexed as an immutable diagnostic with a per-raffle counter; production alerting should also consume finalized raw logs `[RF-035]`.
6. `totalEntries` and `ticketCount` are independent counters. Keep both, and keep both as `bigint` `[RF-017]`.
7. Ticket burns emit ERC-721 `Transfer` to the zero address and are the only ticket destruction path.
8. Each raffle captures its own `protocolTreasury` immutably; read the raffle, not the factory.

14.5 Gas characteristics

Purchase is one payment plus one mint regardless of entry count; the callback is one modulo plus bounded writes; winner proof is one range load `[RF-063]`. Regressions assert that callback gas does not scale with ticket count, that both terminal branches stay below the 300,000-unit budget, and that purchase and refund gas do not scale with the entries inside a ticket. Refund cost scales with batch length up to the 100-ticket bound. Ticket transfers to code-bearing addresses incur one external `isRaffle` call `[RF-025]`. The committed deterministic gas snapshot is the reference; re-measure on the release SHA.

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15. Deployment and operational requirements

Full procedure: `docs/DEPLOYMENT.md`. Executable gate list: `packages/contracts/audit/RELEASE-CHECKLIST.md`.

15.1 Current state

`deployments/` contains only `schema.json`, and `packages/config/src/deployments.ts` exports an **empty** record map, so `protocolIsConfigured === false` in the web app and all writes are disabled. There is no default mainnet deployment command `[RF-065]`.

15.2 Deployment-time verification (must pass before any record is published)

- Official Ethereum chain ID and the current Chainlink VRF v2.5 native wrapper address verified from primary sources **on release day**; wrapper and coordinator code and configuration live-checked.
- Consumer callback limit + wrapper overhead + EIP-150 compensation confirmed to fit the live coordinator maximum.
- Official USDC address verified; six decimals; unpaused; runtime code present; issuer control surface reviewed.
- Treasury is a reviewed contract wallet, code-bearing, and not a predicted clone address.
- Verified published source matching deployed runtime byte-for-byte, with `Proxy === "0"`, an exactly empty `Implementation` field, and no similar-match address (`V1-DEPLOY-02`).
- Factory ABI confirmed ownerless — no owner, role, pause, upgrade, or setter selector.
- Canonical ERC-1167 clone target compared against the deployed implementation at a finalized block.

15.3 Monitoring surface

Creation events; purchases; draw requests and `DrawRequested` fee/excess fields; the two deadline classes; `VrfCallbackIgnored` (from finalized raw logs, not only the subgraph); `RaffleResolved`; `WinningTicketSettled` and `WinningTicketRedeemed`; refund enablement and `remainingRefundLiability` drawdown; the §8.2 solvency inequality per raffle; USDC pause/blocklist indicators; prize-collection pause/upgrade indicators; and sponsor/treasury release events.

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16. Open problems and limitations

16.1 Open problems

#	PROBLEM	STATUS
1	Prize collection can permanently block a final NFT result	**Accepted, most material.** No post-result refund exists `[RF-068]`
2	Claims assigned to future code-less clone addresses	Accepted unsupported; the previous fix was itself exploitable `[RF-072]`
3	Arbitrary non-callable ticket and payout destinations	Deliberately unsolved; undecidable from bytecode `[RF-072]`
4	Modulo bias	Accepted, negligible, documented `[RF-075]`
5	No onchain economic value ceiling	Accepted; requires an explicit launch-policy decision `[RF-073]`
6	Immutability removes every remediation lever	Accepted consequence of G1 `[RF-074]`
7	Requester must fund the VRF fee	Economic assumption; no protocol keeper `[RF-030]`
8	Deployer misconfiguration can poison a factory	Deployment control, not runtime `[RF-071]`
9	Independent audit	**Not performed** `[RF-066]`
10	RPC-backed fork validation for this candidate	**Not performed** — both cases skipped `[RF-067]`
11	External-fuzzer runtime campaign	**Not performed** — harness compiles only `[RF-067]`
12	Monitored Sepolia soak and incident drill	**Not performed** `[RF-079]`
13	Jurisdiction-specific legal review	**Not performed** `[RF-078]`
14	Production treasury Safe selection and review	**Not performed** `[RF-079]`

16.2 What this design retired

Historical audit reports in this repository still describe the following. None of it exists in the current source, and none of it may be reintroduced into a public document from those reports.

RETIRED	CURRENT
Pyth Entropy v2 randomness	Chainlink VRF v2.5 native direct funding (§6)
Base / Base Sepolia target chain	Ethereum mainnet / Sepolia (§0)
`RaffleLens` read aggregator	Nothing. Read raffles directly (§14.1)
`Ownable2Step` factory, treasury setter, creation pause	An ownerless factory with no admin surface (§10.2)
One `CREATE` deployment per raffle	Fixed-target ERC-1167 clones (§2)
Variable `ticketPrice`, `minimumTickets`	Fixed 1 USDC `ENTRY_PRICE`, `reserveEntries` (§7.1)
One ERC-721 per ticket, `1..100` per purchase	One range ticket per purchase, any positive count (§5.1)
`Closed` status for zero-sale raffles	Zero-liability `Refunding` (§9.1)
Transfer lock during `Drawing` and on the winning ticket	No lock in any status (§5.3)
`MAX_START_DELAY` scheduled starts	Sale begins at creation (§3.4)
3-day `DRAW_REQUEST_GRACE_PERIOD`	2-day `DRAW_REQUEST_TIMEOUT` (§3.4)
30-day `NFT_REDEMPTION_TIMEOUT` and `NftWon → Refunding`	Nothing. A resolved result is final (§9.5)
Claims credited inside the oracle callback	Separate permissionless `settleWinningTicket` (§7.4)

RETIRED	CURRENT
Cash split of 80/20 on a post-fee pot	80/5/15 of **gross** (§7.1)
`recoverProtocolOwnedClaim` recovery dispatcher	Nothing. It was exploitable and was removed (§5.4)
`claimQuote` / `claimQuoteFor` pull ledger	Fixed-recipient release functions (§10.1)
`sponsorPrizeRecoveryRecipient` as a distinct role	The single immutable `sponsorRecipient` (§7.2)
`metadataURI` and `MAX_METADATA_URI_LENGTH`	Nothing. No onchain metadata parameter
309-byte EIP-170 headroom on the factory	20,603 bytes of runtime margin (§2.3)

Appendix A — Constants

CONSTANT	VALUE
`BPS`	10,000
`PROTOCOL_FEE_BPS`	500
`CASH_WINNER_BPS`	8,000
`ENTRY_PRICE`	1,000,000
`QUOTE_TOKEN_DECIMALS`	6
`MAX_REFUND_TICKET_BATCH_SIZE`	100
`MAX_SALE_DURATION`	30 days
`DRAW_REQUEST_TIMEOUT`	2 days
`DRAW_CALLBACK_TIMEOUT`	2 days
`VRF_CALLBACK_GAS_LIMIT`	300,000
`VRF_REQUEST_CONFIRMATIONS`	30

Appendix B — External function reference

B.1 `Raffle`

SIGNATURE	MUTABILITY	AUTHORIZATION
`initialize(RaffleInitParams)`	nonpayable	factory only, once
`buyEntries(address,uint128) → uint256`	nonpayable	any; status and window gated
`requestDraw() → uint256`	**payable**	any; window gated
`enableRefunds()`	nonpayable	any at the deadline; sponsor early if unsold
`settleWinningTicket(uint256) → uint256`	nonpayable	**any**; ownership not read
`redeemWinningTicket(uint256) → uint256`	nonpayable	`ownerOf(ticketId)` only
`refundTickets(uint256[]) → uint256`	nonpayable	owner of every listed ticket
`releaseSponsorProceeds() → uint256`	nonpayable	any; pays `sponsorRecipient`
`releaseProtocolFees() → uint256`	nonpayable	any; pays `protocolTreasury`

SIGNATURE	MUTABILITY	AUTHORIZATION
`releaseSponsorPrize()`	nonpayable	any; pays `sponsorRecipient`
`getVrfRequestPrice() → uint256`	view	—
`estimateVrfRequestPrice(uint256) → uint256`	view	—
`ticketRange(uint256) → (uint128,uint128)`	view	historical metadata, survives burn
`drawRequestDeadline() → uint256`	view	—
`callbackDeadline() → uint256`	view	0 before a request
`accountedQuoteBalance() → uint256`	view	—
`grossSales() → uint256`	view	derived, not stored
`onERC721Received(address,address,uint256,bytes) → bytes4`	nonpayable	exact prize only
`receive()`	payable	**always reverts**

Plus the ERC-721 surface and all public state getters declared in `IRaffle`. `name()` is `"raffle.fun Ticket"`; `symbol()` is `"RAFFLE"`.

B.2 `RaffleFactory`

SIGNATURE	AUTHORIZATION
`createRaffle(CreateRaffleParams) → address`	**any, always**
`quoteToken()`, `vrfWrapper()`, `protocolTreasury()`, `raffleImplementation()`	view, immutable
`callbackGasLimit()`, `requestConfirmations()`	view, constant
`raffleCount()`, `isRaffle(address)`, `raffleById(uint256)`, `idByRaffle(address)`	view

There is no owner, setter, pause, upgrade, or rescue function.

Appendix C — Event reference

`Raffle`

`PrizeDeposited(address indexed prizeToken, uint256 indexed prizeTokenId, address indexed sponsor)` · `TicketPurchased(address indexed buyer, address indexed recipient, uint256 indexed ticketId, uint128 firstEntry, uint128 lastEntry, uint128 entryCount, uint256 grossAmount)` · `DrawRequested(uint256 indexed requestId, address indexed requester, uint256 fee, uint256 excessReturned, uint256 drawRequestedAt, uint256 callbackDeadline)` · `VrfCallbackIgnored(uint256 indexed receivedRequestId, uint256 indexed expectedRequestId, Status status)` · `RaffleResolved(uint256 indexed requestId, uint128 indexed winningEntry, Status indexed result)` · `RefundsEnabled(address indexed finalizer, uint256 remainingRefundLiability)` · `WinningTicketSettled(uint256 indexed ticketId, address indexed settler, Status indexed result, uint256 cashAmount, uint256 protocolFee, uint256 sponsorAmount)` · `WinningTicketRedeemed(uint256 indexed ticketId, address indexed winner, Status indexed result, uint256 cashAmount, address prizeToken, uint256 prizeTokenId)` · `TicketsRefunded(address indexed owner, uint256 ticketQuantity, uint256 entryQuantity, uint256 amount, uint256 remainingRefundLiability)` · `SponsorProceedsReleased(address indexed caller, address indexed recipient, uint256 amount)` · `ProtocolFeesReleased(address indexed caller, address indexed treasury, uint256 amount)` · `SponsorPrizeReleased(address indexed caller, address indexed recipient, address indexed prizeToken, uint256 prizeTokenId)`

`RaffleFactory`

`RaffleCreated(uint256 indexed raffleId, address indexed raffle, address indexed sponsor, address sponsorRecipient, address prizeToken, uint256 prizeTokenId, address quoteToken, address protocolTreasury, uint128 reserveEntries, uint64 endTime)`

This is the only event the factory emits, and it is indexer-complete: an indexer never needs to call back into the factory to reconstruct a raffle's fixed configuration.

Appendix D — Error reference

`IRaffle` — `OnlyFactory`, `AlreadyInitialized`, `ZeroAddress`, `InvalidStatus`, `UnexpectedPrize`, `SaleEnded`, `InvalidRecipient`, `ZeroEntryCount`, `TotalEntriesOverflow`, `InvalidTicketBatchSize`, `UnsupportedQuoteToken`, `UnsupportedQuoteTokenTransfer`, `InvalidQuoteDestination`, `OnlySponsor`, `RaffleNotEnded`, `DrawRequestWindowExpired`, `NoEntriesSold`, `RefundsNotAvailable`, `InsufficientVrfFee`, `NativeRefundFailed`, `NoWinnerProceeds`, `NoSponsorProceeds`, `NoProtocolFees`, `UnsafeProtocolDestination`, `NotTicketOwner`, `TicketDoesNotContainWinningEntry`, `SettlementAlreadyComplete`, `WinningTicketMismatch`, `WinningTicketAlreadyRedeemed`, `SponsorPrizeUnavailable`, `PrizeAlreadyClaimed`, `PrizeDeliveryVerificationFailed`.

`Raffle` (local) — `DirectNativeTransfer`.

`IRaffleFactory` — `ZeroAddress`, `NotContract`, `UnsupportedQuoteToken`, `InvalidQuoteTokenDecimals`, `UnsupportedPrizeToken`, `UnsafeProtocolDestination`, `ZeroReserveEntries`, `InvalidEndTime`, `SaleDurationTooLong`, `PrizeEscrowVerificationFailed`.

Appendix E — Primary sources

TOPIC	SOURCE
Contracts	`packages/contracts/src/{Raffle,RaffleFactory}.sol`, `src/interfaces/`, `src/libraries/RaffleConstants.sol`
Fact registry	`docs/facts/raffle-fun-facts.md`
Architecture	`docs/ARCHITECTURE.md`
Lifecycle	`docs/STATE-MACHINE.md`
Economics	`docs/ECONOMICS.md`
Randomness	`docs/RANDOMNESS.md`
Threat model	`docs/THREAT-MODEL.md`
Invariants	`docs/SECURITY-INVARIANTS.md`
Deployment	`docs/DEPLOYMENT.md`
Monitoring	`docs/MONITORING.md`
Incident response	`docs/INCIDENT-RESPONSE.md`
Current specification	`packages/contracts/audit/CURRENT-SPECIFICATION.md`
Current campaign	`packages/contracts/audit/CURRENT-CAMPAIGN.md`, `CURRENT-TEST-MATRIX.md`
Current findings	`packages/contracts/audit/CURRENT-FINDINGS.md`, `CURRENT-RESIDUAL-RISKS.md`
Latest review	`packages/contracts/audit/V12-REVIEW-2026-08-20.md`

TOPIC	SOURCE
Release gates	`packages/contracts/audit/RELEASE-CHECKLIST.md`, `RELEASE-READINESS-2026-08-18.md`
Disclosure policy	`SECURITY.md`
Chainlink VRF v2.5	https://docs.chain.link/vrf
USDC addresses	https://developers.circle.com/stablecoins/usdc-contract-addresses

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